

NVE Introduces World's Most Sensitive Magnetometer IC & More Ultra-Miniature TMR Converters

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Industry Announcement & Technological Breakthrough

Billed as the world's most sensitive magnetometer integrated circuit, the ALT021-10E analog magnetometer from NVE Corporation demonstrates an unprecedented sensitivity of 0.5 millivolts per volt per microtesla (0.5 mV/V/ μ T). Engineered utilizing NVE Corporation's proprietary Tunneling Magnetoresistance (TMR) spintronic technology, the newly launched ALT-Series sensors deliver high-precision magnetic field detection from fractions of a microtesla up to 10 millitesla.

ALT-Series Portfolio Comparison & Field Specifications

Sensor Part Number	Linear Magnetic Field Range	Nominal Bridge Resistance	Typical Application Focus
ALT021-10E	± 0.25 mT (± 250 μ T)	20 k Ω $\pm$ 20%	Ultra-low field sensing, geomagnetic navigation, biomagnetic detection
ALT023-10E	± 1.0 mT (± 10 Oe)	20 k Ω $\pm$ 20%	Precision magnetic biosensors, Helmholtz instrumentation, eddy-current testing
ALT025-10E	± 10.0 mT (± 100 Oe)	20 k Ω $\pm$ 20%	Motor position feedback, high-current proximity detection, industrial encoders

Physical Architecture & Packaging Advantages

All sensors in the ALT series are packaged in a miniature 1.1 mm $\times$ 1.1 mm $\times$ 0.45 mm 4-pin Ultra-DFN (DFN4) package, occupying a minuscule footprint on printed circuit boards. Each device incorporates a fully integrated four-element Wheatstone bridge with high TMR ratio ($>100\%$), exhibiting minimal 1/f flicker noise and exceptionally low power consumption. For laboratory and instrumentation prototyping, NVE Corporation also supplies the EVB01 Evaluation Board, featuring an on-board ALT023-10E sensor with decoupling capacitors and SMA/test-point breakouts for immediate coupling to instrumentation amplifiers such as the AD623.

Relevance to Nanofiber Magnetic Biosensing

In biosensor platforms utilizing superparamagnetic iron oxide nanoparticles (Fe₃O₄), the ALT023-10E sensor operates in its high-linearity ± 1.0 mT regime under external Helmholtz coil excitation. Changes in local magnetic dipole flux resulting from target analyte binding (e.g. formaldehyde reacting with ADH receptor sites on electrospun Fe₃O₄/PVA nanofibers) induce differential voltage variations at the bridge terminals, which are subsequently amplified by low-noise signal conditioning circuitry.